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Master of Computer Applications

Semester – III

Data Analytics for Business

Experiment No.: 6

Title: Creating HR Analytics Dashboards

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Experiment No: 06

Title: Creating HR Analytics Dashboards

Problem Statement

You are an HR Data Analyst in a multinational organization. The HR department wants to analyze employee demographics, workforce distribution, attrition patterns, and employee-related factors using visual reports. The objective is to create an interactive HR analytics dashboard in Power BI using the IBM HR Analytics Employee Attrition dataset so that HR management can easily understand workforce characteristics, identify attrition patterns, and make better employee management and workforce planning decisions.

Methodology / Steps Followed

Step 1: Data Collection

- Downloaded the IBM HR Analytics Employee Attrition Dataset from Kaggle.
- The dataset contains employee-related information such as demographics, job details, compensation, satisfaction, experience, and attrition.
- The dataset was selected to analyze employee workforce and attrition patterns.

Step 2: Data Import

- Opened **Power BI Desktop**.
- Clicked **Get Data** → **Text/CSV**.
- Selected the IBM HR Analytics dataset.
- Loaded the dataset into Power BI.
- Verified that the required columns were imported correctly.

Step 3: Data Cleaning and Transformation

Using Power Query:

- Checked for missing/null values.
- Checked for duplicate employee records.
- Verified the data types of all columns.
- Removed unnecessary columns if required.
- Ensured categorical fields such as Department, Gender, Job Role, and Attrition were correctly formatted.

- Created suitable Age Groups for demographic analysis.
- Created Years-at-Company Groups for workforce experience analysis.
- Verified the final dataset before creating visualizations.

KPI Cards Analysis



1. Total Employees

- Represents the total number of employees in the dataset.
- Helps HR understand the overall workforce size.

2. Total Attrition

- Represents the number of employees who have left the organization.
- Helps HR monitor employee turnover.

3. Attrition Rate

- Represents the percentage of employees who left the organization compared with the total workforce.
- Helps HR evaluate the overall employee retention situation.

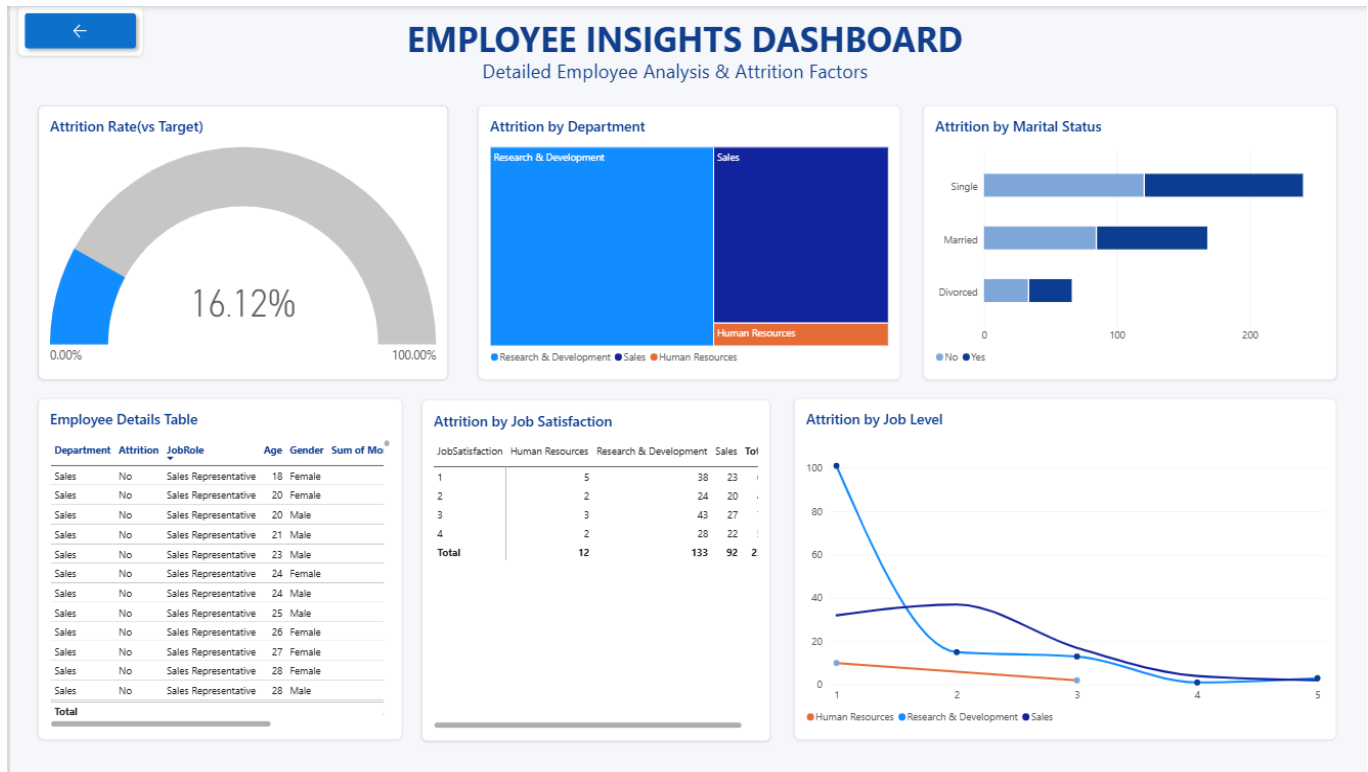
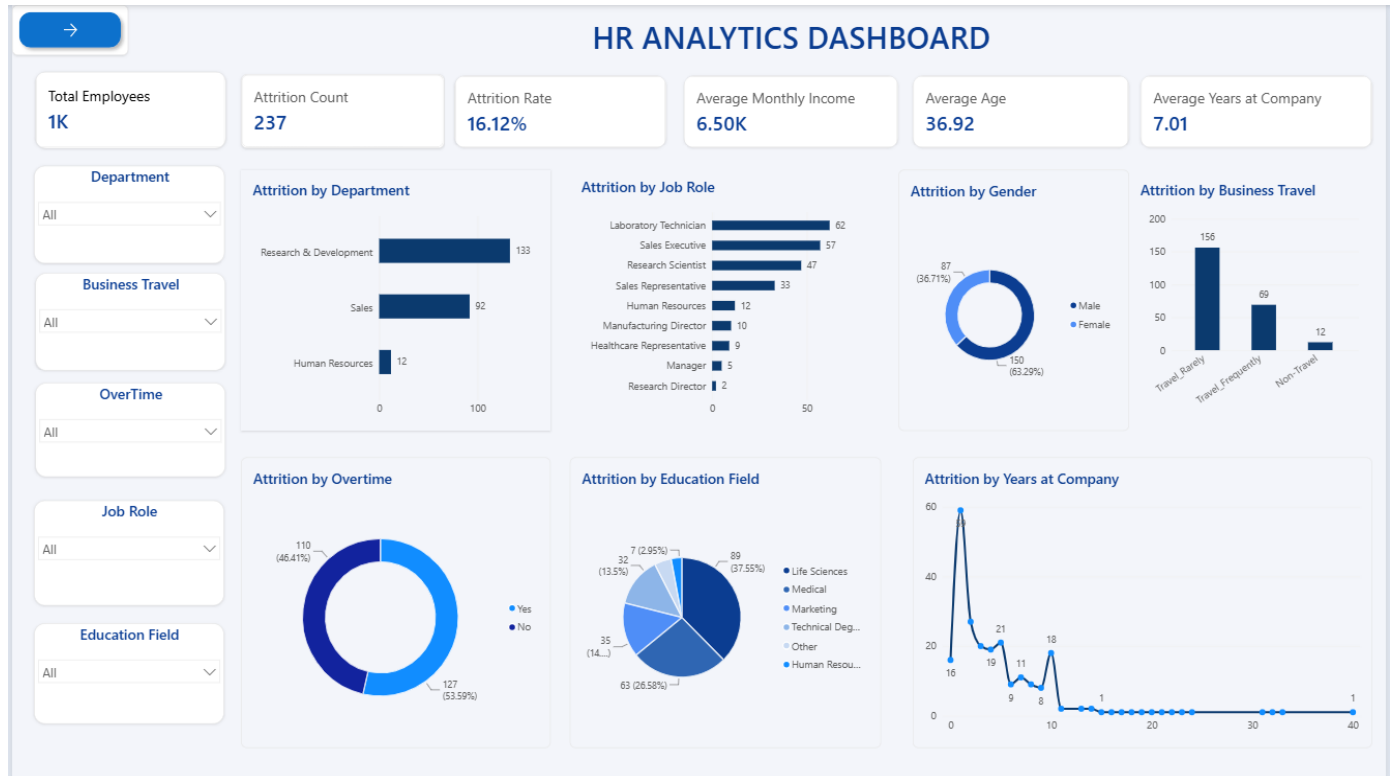
4. Average Age

- Represents the average age of employees.
- Helps HR understand workforce demographics.

5. Average Monthly Income

- Represents the average employee income.
- Helps HR analyze compensation patterns.

Visualizations Analysis



1. Attrition by Department — Bar Chart

Purpose: Shows the number of employees who left the organization across different departments.

Observation:

- Research & Development has the highest attrition count with 133 employees.
- Sales follows with 92 employees.
- Human Resources has the lowest attrition count with 12 employees.

Business Insight:

- Research & Development requires greater attention from HR because it has the highest number of employee exits.
- Retention strategies can be prioritized for high-attrition departments.

2. Attrition by Job Role — Bar Chart

Purpose: Compares employee attrition across different job roles.

Observation:

- Laboratory Technicians have the highest attrition count with 62 employees.
- Sales Executives follow with 57.
- Research Scientists have 47, while Sales Representatives have 33.
- Other roles have comparatively lower attrition.

Business Insight:

- Laboratory Technician and Sales Executive roles show relatively high employee turnover.
- HR can investigate workload, career growth, compensation, and job satisfaction for these roles.

3. Attrition by Gender — Donut Chart

Purpose: Shows the distribution of attrition by gender.

Observation:

- Male employees account for 150 attrition cases (63.29%).
- Female employees account for 87 cases (36.71%).

Business Insight:

- The dashboard indicates a higher number of attrition cases among male employees.

4. Attrition by Business Travel — Column Chart

Purpose: Analyzes attrition according to business travel frequency.

Observation:

- Travel_Rarely has the highest attrition count with 156 employees.
- Travel_Frequently accounts for 69 employees.
- **Non-Travel** accounts for 12 employees.

Business Insight:

- Employees who travel rarely represent the largest group of attrition cases.
- HR should consider business travel together with other factors rather than assuming frequent travel is the primary reason for attrition.

5. Attrition by Overtime — Donut Chart

Purpose: Compares attrition between employees who work overtime and those who do not.

Observation:

- Employees working overtime account for **127 attrition cases (53.59%)**.
- Employees not working overtime account for **110 cases (46.41%)**.

Business Insight:

- Attrition is slightly higher among employees who work overtime.
- HR should monitor workload and work-life balance and consider appropriate workload-management strategies.

6. Attrition by Education Field — Donut Chart

Purpose: Shows attrition distribution across different educational backgrounds.

Observation:

- Life Sciences has the highest count with 89 cases (37.55%).
- Medical follows with 63 cases (26.58%).
- Marketing accounts for 35 cases.
- Technical Degree accounts for 32 cases.
- Other education fields have comparatively smaller numbers.

Business Insight:

- Employees from Life Sciences and Medical backgrounds represent a significant proportion

of attrition.

- HR can examine whether these employees are concentrated in particular departments or job roles.

7. Attrition by Years at Company — Line Chart

Purpose: Analyzes attrition according to employee tenure.

Observation:

- Attrition is highest among employees with 0 years at the company, with approximately 59 cases.
- Attrition decreases considerably as years at the company increase.
- There are smaller increases around 1, 2, 3, and 5 years.
- Attrition becomes very low for employees with longer tenure.

Business Insight:

- Employees in their early stage of employment appear to be more vulnerable to leaving.
- HR should strengthen onboarding, employee engagement, career development, and early-stage retention programs.

8. Attrition Rate vs Target — Gauge Chart

Purpose: Displays the overall employee attrition rate.

Observation:

- The current attrition rate is 16.12%.
- The gauge provides a simple view of the overall attrition level.

Business Insight:

- An attrition rate of 16.12% indicates that employee turnover is an important area for HR monitoring.
- The organization can use this KPI to track improvement in future periods and evaluate the effectiveness of retention strategies.

9. Attrition by Marital Status — Stacked Bar Chart

Purpose: Compares attrition between different marital-status groups.

Observation:

- Single employees represent the largest workforce group in the visualization.
- Married employees form the next major group.

- Divorced employees represent the smallest group.
- The chart distinguishes between employees who stayed and those who left.

Business Insight:

- Marital status can be considered as one of the demographic factors when analyzing employee attrition.
- HR should combine this information with age, job role, department, and other factors before making retention decisions.

10. Employee Details Table

Purpose: Provides detailed employee-level information.

Displayed information includes:

- Department
- Attrition
- Job Role
- Age
- Gender
- Monthly Income

Business Insight:

- The table allows HR managers to drill down from overall dashboard trends to individual employee records.
- It supports detailed investigation of employees across departments and job roles.

11. Attrition by Job Satisfaction — Matrix

Purpose: Shows the relationship between job satisfaction levels and departments.

Observation:

- Job satisfaction levels from 1 to 4 are displayed.
- The matrix allows comparison between Human Resources, Research & Development, and Sales.
- Research & Development has the largest overall employee count among the displayed departments.

Business Insight:

- HR can use satisfaction levels to identify departments where employee engagement may

require improvement.

- Combining job satisfaction with attrition can help identify potential retention risks.

12. Attrition by Job Level — Line Chart

Purpose: Compares employee distribution/attrition across different job levels.

Observation:

- The chart shows substantial differences between job levels.
- Lower job levels contain a larger number of employees.
- The number decreases as job level increases.

Business Insight:

- Employees at lower job levels represent an important area for retention planning.
- HR can focus on career progression, promotion opportunities, training, and employee development.

Overall HR Insights

- The overall attrition rate is 16.12%, with 237 employees leaving the organization.
- Research & Development has the highest attrition, followed by Sales.
- Laboratory Technicians and Sales Executives have higher attrition among job roles.
- Employees who work overtime show slightly higher attrition.
- Male employees account for a higher share of attrition cases than female employees.
- Attrition is highest among employees with low years at the company, especially new employees.
- Life Sciences has the highest attrition among education fields.
- The dashboard shows that job satisfaction, job level, business travel, and marital status can be considered when analyzing employee attrition.
- HR can use these insights to improve employee retention, engagement, and workforce planning.

Conclusion

The HR Analytics Dashboard helps to understand employee details, workforce distribution, and attrition in a simple way. It highlights departments and job roles with higher attrition and shows factors such as overtime, job satisfaction, gender, and years at the company. These insights can help HR improve employee retention, workforce planning, and employee management.